

Assignment 4

1. Question 1: Print Numbers from 1 to N

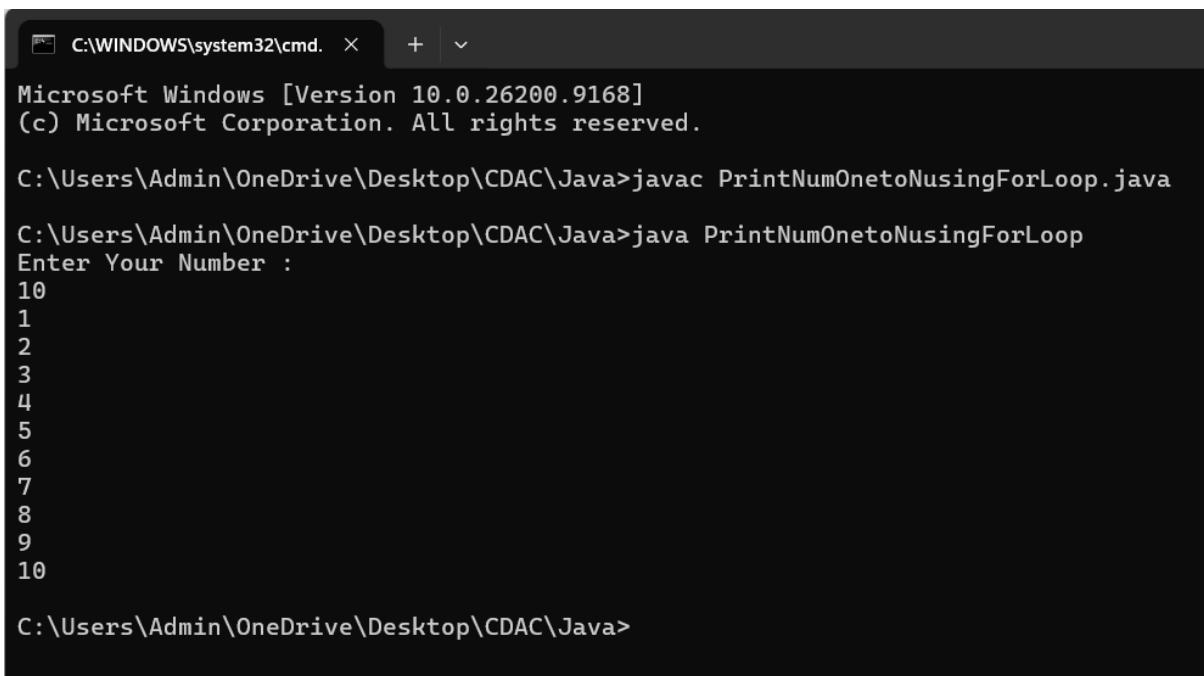
1.1 Code

```
import java.util.Scanner;
public class PrintNumOnetoNusingForLoop
{
    public static void main(String args[])
    {
        Scanner obj = new Scanner(System.in);

        System.out.println("Enter Your Number : ");
        int n = obj.nextInt();

        for(int i = 1; i <= 10; i++)
        {
            System.out.println(i);
        }
    }
}
```

1.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac PrintNumOnetoNusingForLoop.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java PrintNumOnetoNusingForLoop
Enter Your Number :
10
1
2
3
4
5
6
7
8
9
10

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

2. Question 2: Print Multiples of 3 between 1 and N

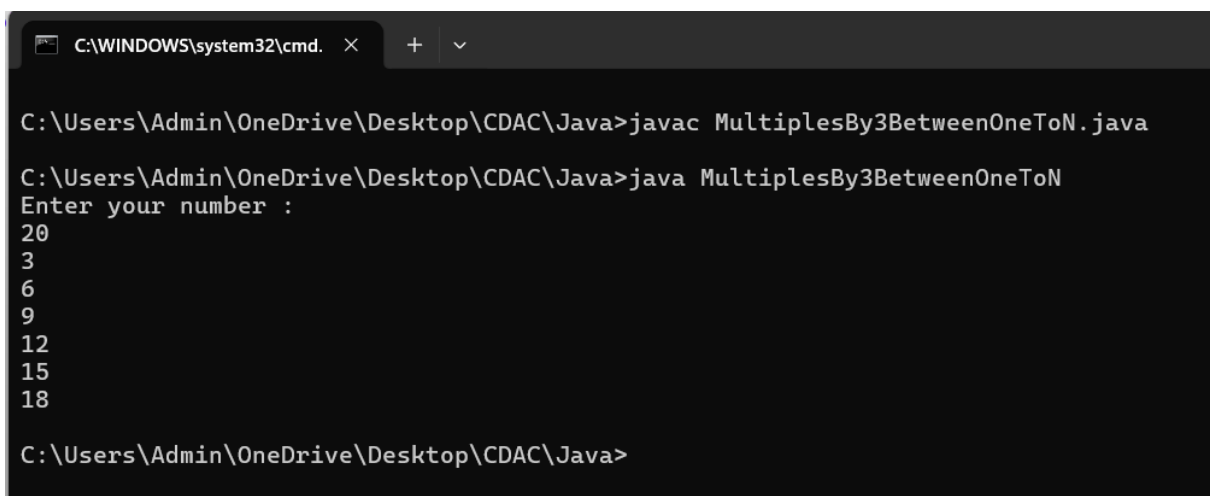
2.1 Code

```
import java.util.Scanner;
public class MultiplesBy3BetweenOneToN
{
    public static void main(String args[])
    {
        Scanner obj = new Scanner(System.in);

        System.out.println("Enter your number : ");
        int n = obj.nextInt();

        for(int i = 3; i <= n; i= i + 3)
        {
            System.out.println(i);
        }
    }
}
```

2.2 Output



```
C:\WINDOWS\system32\cmd.  x  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac MultiplesBy3BetweenOneToN.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java MultiplesBy3BetweenOneToN
Enter your number :
20
3
6
9
12
15
18

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

3. Question 3: Calculate the Factorial of a Number

3.1 Code

```
import java.util.Scanner;

public class FactorialNum
{
    //5 × 4 × 3 × 2 × 1 = 120
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

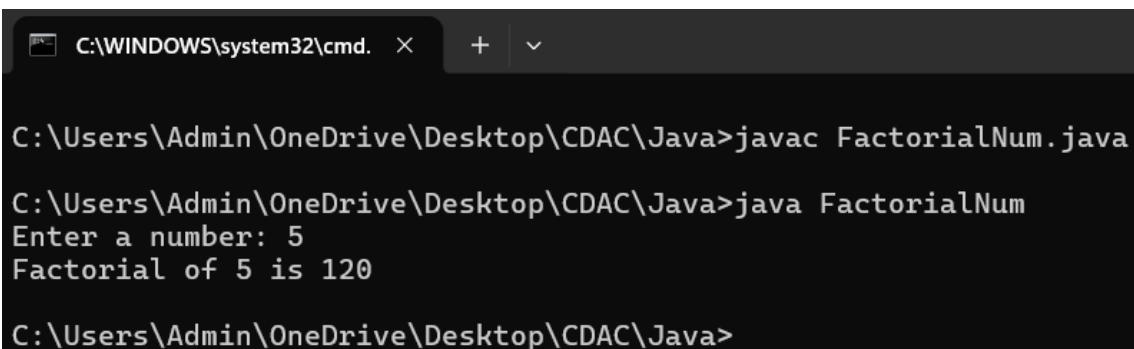
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int factorial = 1;

        for (int i = 1; i <= n; i++)
        {
            factorial = factorial * i;
        }

        System.out.println("Factorial of " + n + " is " + factorial);
    }
}
```

3.2 Output



```
C:\WINDOWS\system32\cmd.  ×  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac FactorialNum.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java FactorialNum
Enter a number: 5
Factorial of 5 is 120

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

4. Question 4: Print Even Numbers from 1 to N

4.1 Code

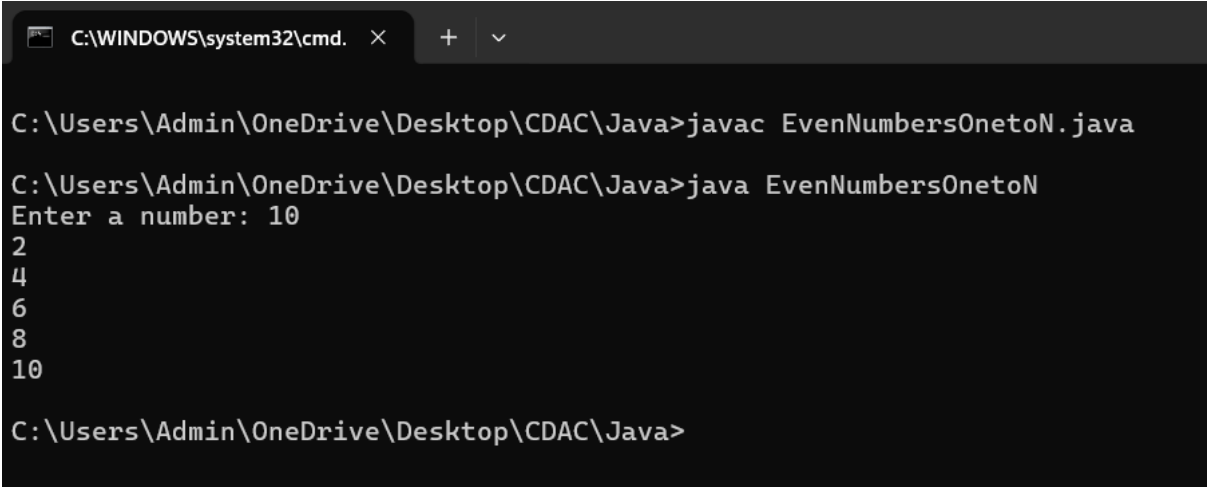
```
import java.util.Scanner;

public class EvenNumbersOnetoN
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        for (int i = 2; i <= n; i = i + 2)
        {
            System.out.println(i);
        }
    }
}
```

4.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac EvenNumbersOnetoN.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java EvenNumbersOnetoN
Enter a number: 10
2
4
6
8
10

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

5. Question 5: Sum of Odd Numbers between 1 and N

5.1 Code

```
import java.util.Scanner;

public class SumOfOddNumbers
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
```

```

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

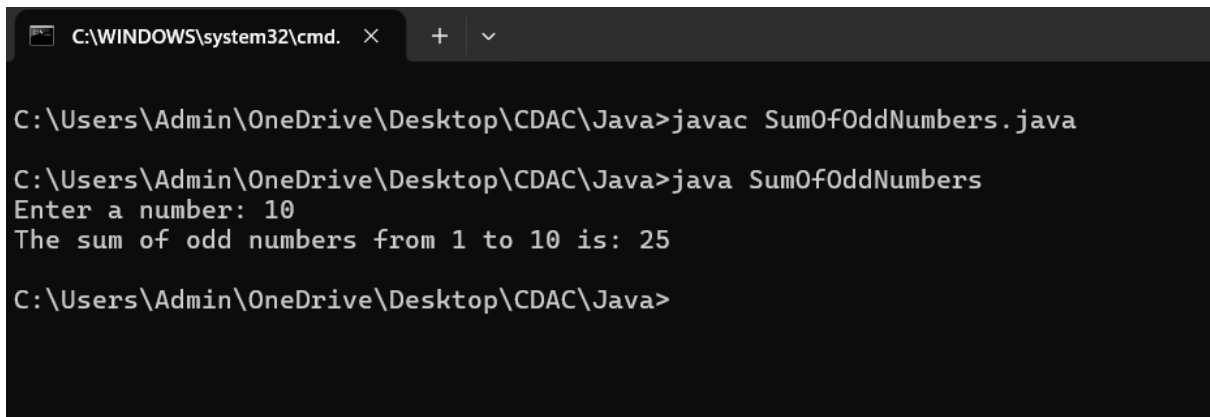
        int sum = 0;

        for (int i = 1; i <= n; i++)
        {
            if (i % 2 != 0)
            {
                sum = sum + i;
            }
        }

        System.out.println("The sum of odd numbers from 1 to " + n + " is: " + sum);
    }
}

```

5.2 Output



```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac SumOfOddNumbers.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java SumOfOddNumbers
Enter a number: 10
The sum of odd numbers from 1 to 10 is: 25

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

6. Question 6: Print All Elements of an Array

6.1 Code

```

import java.util.Scanner;

public class ArrayElementsForEach
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        int numbers[] = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++)
        {

```

```

        numbers[i] = sc.nextInt();
    }

    for (int number : numbers)
    {
        System.out.println(number);
    }
}
}

```

6.2 Output

```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac ArrayElementsForEach.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java ArrayElementsForEach
Enter 5 integers:
4
7
11
12
14

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

7. Question 7: Find the Sum of All Elements in an Array

7.1 Code

```

import java.util.Scanner;
public class ArrayElementOfSum
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        int numbers[] = new int[5];

        System.out.println("Enter Five integers :");

        for (int i = 0; i < 5; i++)
        {
            numbers[i] = sc.nextInt();

```

```

    }

    int sum = 0;

    for (int number : numbers)
    {
        sum = sum + number;
    }

    System.out.println("The sum of all numbers is: " + sum);
}
}

```

7.2 Output

```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac ArrayElementOfSum.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java ArrayElementOfSum
Enter Five integers :
4 7 11 14 17
The sum of all numbers is: 53

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

8. Question 8: Print All Names in a String Array

8.1 Code

```

import java.util.Scanner;

public class AllNamesStringArray
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        String names[] = new String[4];

        System.out.println("Enter Four Names:");

        for (int i = 0; i < 4; i++)

```

```

        {
            names[i] = sc.next();
        }

        for (String name : names)
        {
            System.out.println(name);
        }
    }
}

```

8.2 Output

```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac AllNamesStringArray.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java AllNamesStringArray
Enter Four Names:
abc xyz pqr asd qwe
abc
xyz
pqr
asd

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

9. Question 9: Find the Largest Element in an Array

9.1 Code

```

import java.util.Scanner;
import java.util.Arrays;

public class LargestElementOfArray
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        int numbers[] = new int[5];

        System.out.println("Enter Five integers:");

        for (int i = 0; i < 5; i++)
        {

```



```

        numbers[i] = sc.nextInt();
    }

    Arrays.sort(numbers);

    System.out.println("The largest element is: " + numbers[4]);
}
}
}

```

9.2 Output

```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac LargestElementOfArray.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java LargestElementOfArray
Enter Five integers:
7 11 17 14 8
The largest element is: 17

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

10. Question 10: Find the Average of Elements in an Array

10.1 Code

```

import java.util.Scanner;

public class AvgElementsOfArray
{
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);

        int numbers[] = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++)
        {
            numbers[i] = sc.nextInt();
        }

        int sum = 0;

        for (int number : numbers)
        {

```

```

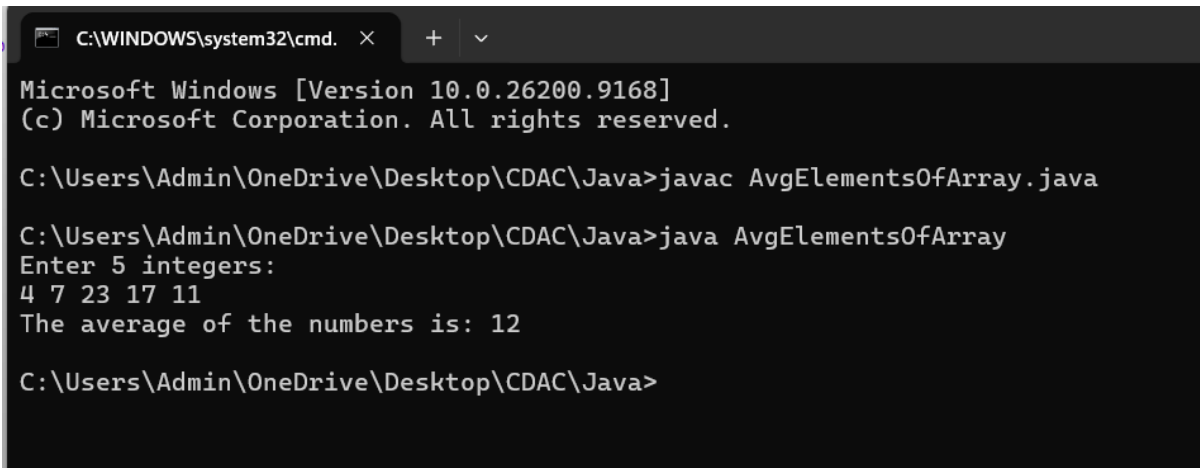
        sum = sum + number;
    }

    int average = sum / 5;

    System.out.println("The average of the numbers is: " + average);
}
}
}

```

10.2 Output



```

C:\WINDOWS\system32\cmd.  X  +  v

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac AvgElementsOfArray.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java AvgElementsOfArray
Enter 5 integers:
4 7 23 17 11
The average of the numbers is: 12

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

11. Question 11: Count Positive and Negative Numbers in an Array

11.1 Code

```

import java.util.Scanner;

public class CountPositivenNegativeNumInArray
{
    public static void main(String args[])
    {
        Scanner obj = new Scanner(System.in);

        int numbers[] = new int[5];

        System.out.println("Enter 6 integers:");

        for (int i = 0; i < 5; i++)
        {
            numbers[i] = obj.nextInt();
        }

        int positive = 0;
        int negative = 0;
    }
}

```

```

        for (int number : numbers)
        {
            if (number > 0)
            {
                positive++;
            }
            else if (number < 0)
            {
                negative++;
            }
        }

        System.out.println("Positive numbers: " + positive);
        System.out.println("Negative numbers: " + negative);
    }
}

```

11.2 Output

```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac CountPositivenNegativeNumInArray.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java CountPositivenNegativeNumInArray
Enter 6 integers:
4 -7 11 -14 23
Positive numbers: 3
Negative numbers: 2

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

12. Question 12: Sort an Array in Ascending Order

12.1 Code

```

import java.util.Scanner;
import java.util.Arrays;

public class ArrayAscendingOrder
{
    public static void main(String args[])
    {
        Scanner obj = new Scanner(System.in);
        int numbers[] = new int[5];
        System.out.println("Enter five Numbers : ");
    }
}

```

```

        for(int i = 0; i < 5; i++)
        {
            numbers[i] = obj.nextInt();
        }
        Arrays.sort(numbers);
        System.out.println("Sorted Array : ");

        for(int number : numbers)
        {
            System.out.println(number);
        }
    }
}

```

12.2 Output

```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac ArrayAscendingOrder.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java ArrayAscendingOrder
Enter five Numbers :
23 14 4 17 7
Sorted Array :
4
7
14
17
23

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

13. Question 13: Check if an Array Contains a Specific Element

13.1 Code

```

import java.util.Scanner;
import java.util.Arrays;

public class ArrayContainsSpecificElement
{
    public static void main(String args[])
    {
        Scanner obj = new Scanner(System.in);

        Integer number[] = new Integer[5];
    }
}

```

```

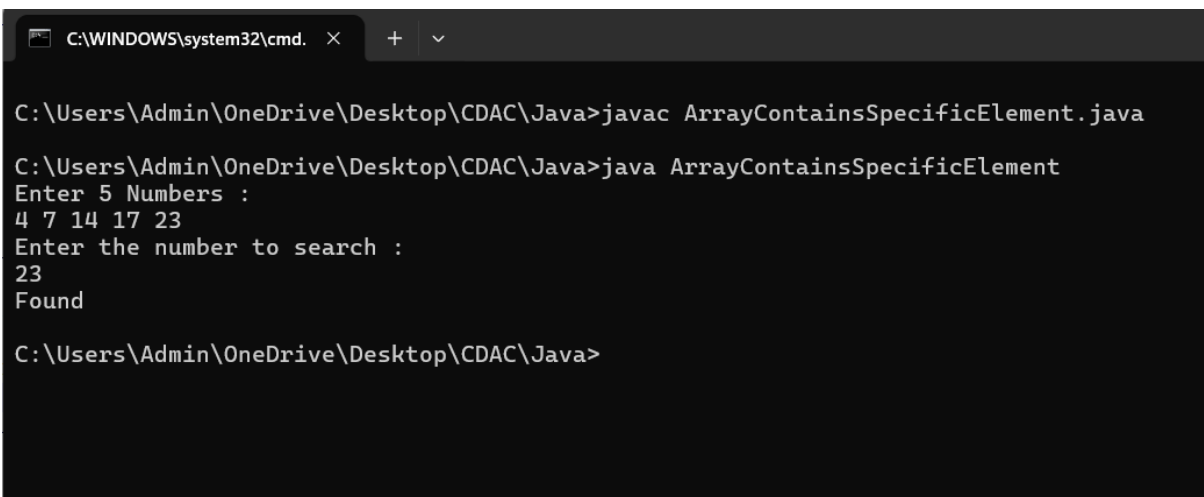
        System.out.println("Enter 5 Numbers :");

        for(int i = 0; i < 5; i++)
        {
            number[i] = obj.nextInt();
        }
        System.out.println("Enter the number to search : ");
        int search = obj.nextInt();

        if (Arrays.asList(number).contains(search))
        {
            System.out.println("Found");
        }
        else
        {
            System.out.println("Not Found");
        }
    }
}

```

13.2 Output



```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac ArrayContainsSpecificElement.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java ArrayContainsSpecificElement
Enter 5 Numbers :
4 7 14 17 23
Enter the number to search :
23
Found

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

14. Question 14: Find the Index of an Element in an Array

14.1 Code

```
import java.util.Scanner;
import java.util.Arrays;

public class IndexOfElementArray
{
    public static void main(String[] args)
    {
        Scanner obj = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        for (int i = 0; i < 5; i++)
        {
            numbers[i] = obj.nextInt();
        }

        Arrays.sort(numbers);

        System.out.print("Enter the number to search: ");
        int search = obj.nextInt();

        int index = Arrays.binarySearch(numbers, search);

        if (index >= 0)
        {
            System.out.println("The number " + search + " is found at index " +
index);
        }
        else
        {
            System.out.println("Not Found");
        }
    }
}
```

14.2 Output

```
C:\WINDOWS\system32\cmd.  ×  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac IndexOfElementArray.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java IndexOfElementArray
Enter 5 integers:
4 7 14 17 23
Enter the number to search: 17
The number 17 is found at index 3

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

15. Question 15:Write a program to print the following pattern:

```
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

15.1 Code

```
public class Pattern15OneToFiveLeftSiderNReverse
{
    public static void main(String[] args)
    {
        for(int i = 1; i <= 5; i++)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(i);

                if(j < i)
                    System.out.print("*");
            }
            System.out.println();
        }

        for(int i = 5; i >= 2; i--)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(i);
```

```

        if(j < i)
            System.out.print("*");
        }
        System.out.println();
    }
}

```

15.2 Output



```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac Pattern15OneToFiveLeftSiderNReverse.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java Pattern15OneToFiveLeftSiderNReverse
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

16. Question 16: Write a program to print the following pattern:

```

1
1*2
1*2*3
1*2*3*4
1*2*3*4*5

```

16.1 Code

```

public class Pattern16OneToFive
{
    public static void main(String[] args)
    {
        for(int i = 1; i <= 5; i++)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(j);

                if(j < i)

```



```

        System.out.print("*");
    }
    System.out.println();
}
}
}

```

16.2 Output

```

C:\WINDOWS\system32\cmd.  ×  +  ▾

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac Pattern16OneToFive.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java Pattern16OneToFive
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

17. Question 17: Write a program to print the following pattern:

```

1
1*3
1*3*5
1*3*5*7
1*3*5*7*9

```

17.1 Code

```

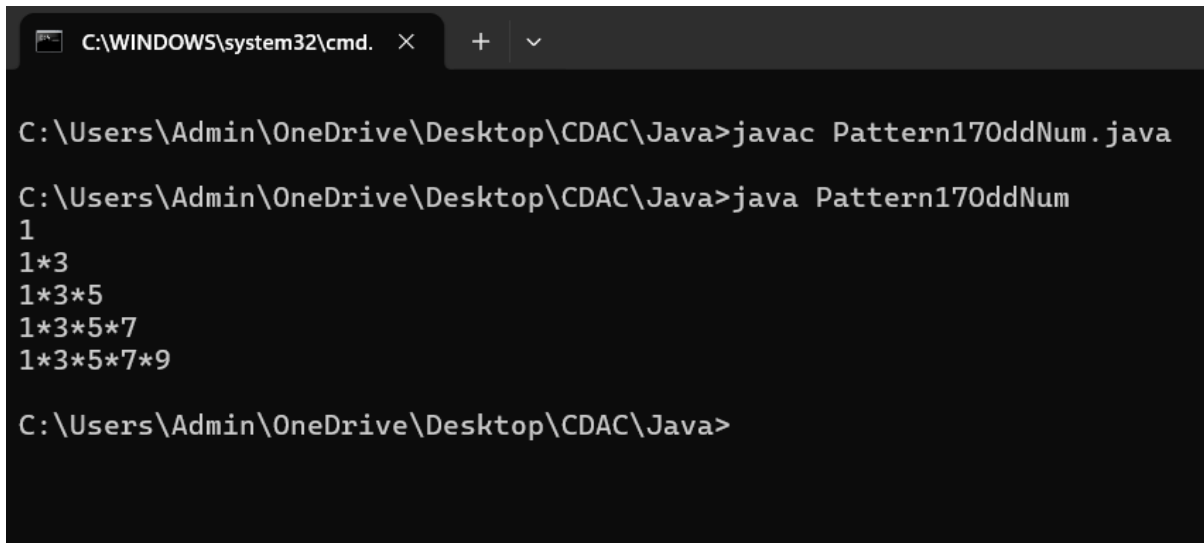
public class Pattern17OddNum
{
    public static void main(String[] args)
    {
        for(int i = 1; i <= 5; i++)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(2 * j - 1);

                if(j < i)
                    System.out.print("*");
            }
            System.out.println();
        }
    }
}

```

```
    }  
  }  
}
```

17.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac Pattern17OddNum.java  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java Pattern17OddNum  
1  
1*3  
1*3*5  
1*3*5*7  
1*3*5*7*9  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

18. Question 18: Write a program to print the following pattern:

```
11111  
22222  
33333  
44444  
55555
```

18.1 Code

```
public class Pattern18Square  
{  
    public static void main(String[] args)  
    {  
        for(int i = 1; i <= 5; i++)  
        {  
            for(int j = 1; j <= 5; j++)  
            {  
                System.out.print(i);  
            }  
  
            System.out.println();  
        }  
    }  
}
```

18.2 Output

```
C:\WINDOWS\system32\cmd. x + v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac Pattern18Square.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java Pattern18Square
11111
22222
33333
44444
55555

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

19. Question 19: Write a program to print the following pattern:

```
1
22
333
4444
55555
```

19.1 Code

```
public class PatternLeftTriangle
{
    public static void main(String args[])
    {
        for(int i = 1; i <= 5; i++)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

19.2 Output

```
C:\WINDOWS\system32\cmd. X + v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac PatternLeftTriangle.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java PatternLeftTriangle
1
22
333
4444
55555

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

20. Question 20: Write a program to print the following pattern:

```
1
12
123
1234
12345
```

20.1 Code

```
public class PatternLeftTriangleNumbers20
{
    public static void main(String args[])
    {
        for(int i = 1; i <= 5; i++)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(j);
            }
            System.out.println();
        }
    }
}
```

20.2 Output

```
C:\WINDOWS\system32\cmd.  ×  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac PatternLeftTriangleNumbers20.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java PatternLeftTriangleNumbers20
1
12
123
1234
12345

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

21. Question 21: Write a program to print the following pattern:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

21.1 Code

```
public class Pattern21OneTo15
{
    public static void main(String[] args)
    {
        int num = 1;

        for(int i = 1; i <= 5; i++)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(num + " ");
                num++;
            }

            System.out.println();
        }
    }
}
```

21.2 Output

```
C:\WINDOWS\system32\cmd. x + v
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac Pattern21OneTo15.java
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java Pattern21OneTo15
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

22. Question 22: Write a program to print the following pattern:

```
*****
*      *
*      *
*      *
*      *
*      *
*****
```

22.1 Code

```
public class Pattern22SquareBox
{
    public static void main(String args[])
    {
        for(int i = 1; i <= 5; i++)
        {
            for(int j = 1; j <= 5; j++)
            {
                if(i == 1 || i == 5 || j == 1 || j == 5)
                {
                    System.out.print("*");
                }else
                {
                    System.out.print(" ");
                }
            }

            System.out.println();
        }
    }
}
```

```

        }
    }
}

```

22.2 Output

```

C:\WINDOWS\system32\cmd. X + v
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac Pattern22SquareBox.java
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java Pattern22SquareBox
*****
*   *
*   *
*   *
*****
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```

23. Question 23: Write a program to print the following pattern:

```

      *
     ***
    *****
   *
  *
 *

```

23.1 Code

```

public class PatternDiamond
{
    public static void main(String[] args)
    {
        for(int i = 1; i <= 7; i++)
        {
            for(int j = 1; j <= 7 - i; j++)
            {
                System.out.print(" ");
            }

            for(int j = 1; j <= 2 * i - 1; j++)
            {
                System.out.print("*");
            }
        }
    }
}

```

```

        System.out.println();
    }

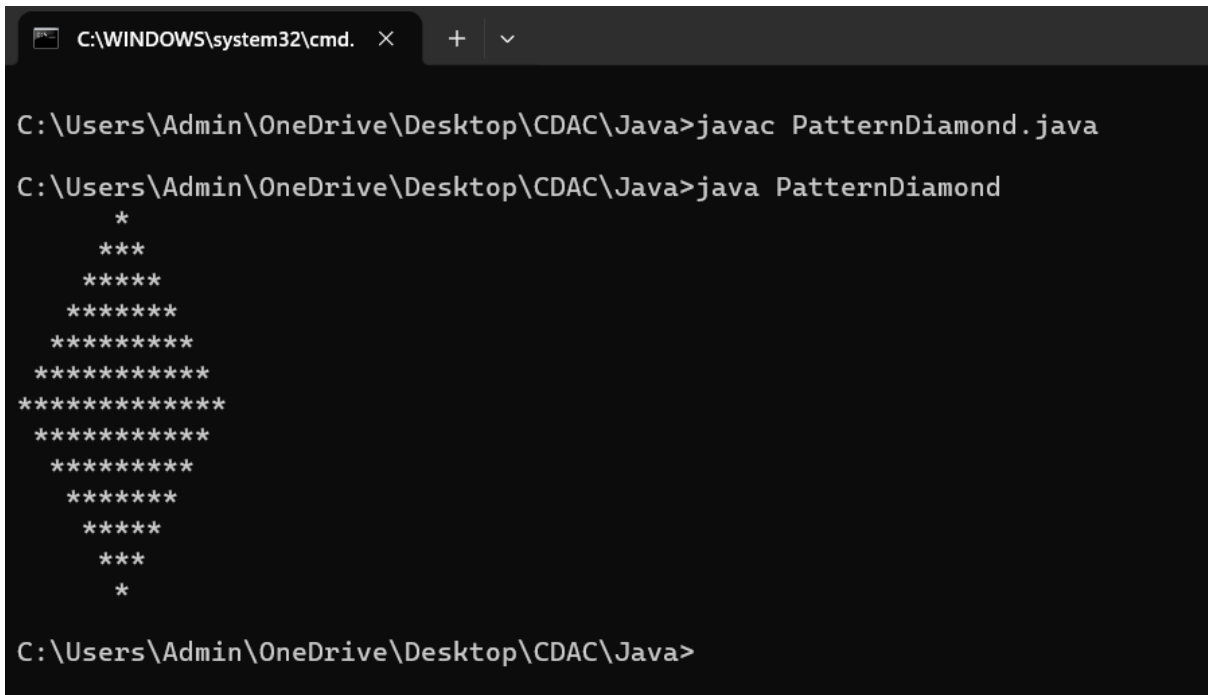
    for(int i = 6; i >= 1; i--)
    {
        for(int j = 1; j <= 7 - i; j++)
        {
            System.out.print(" ");
        }

        for(int j = 1; j <= 2 * i - 1; j++)
        {
            System.out.print("*");
        }

        System.out.println();
    }
}

```

23.2 Output



The screenshot shows a Windows command prompt window with the following content:

```

C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac PatternDiamond.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java PatternDiamond

    *
   ***
  *****
 *****
*****
*****
*****
*****
 *****
  *****
   *****
    *

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>

```


24. Question 24: Reverse a String

24.1 Code

```
import java.util.Scanner;

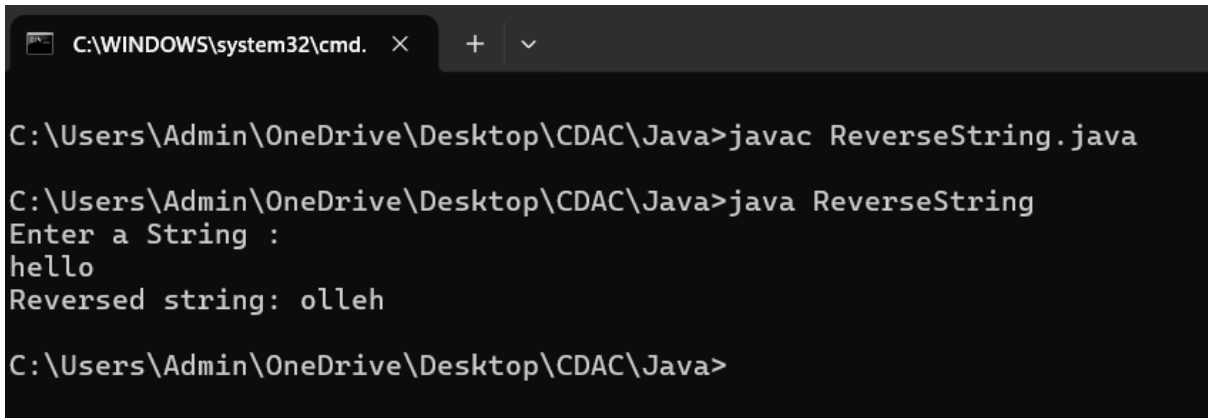
public class ReverseString
{
    public static void main(String args[])
    {
        Scanner obj = new Scanner(System.in);

        System.out.println("Enter a String : ");
        String str = obj.nextLine();

        String reverse = "";
        for (int i = str.length() - 1; i >= 0; i--)
        {
            reverse = reverse + str.charAt(i);
        }

        System.out.println("Reversed string: " + reverse);
    }
}
```

24.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac ReverseString.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java ReverseString
Enter a String :
hello
Reversed string: olleh

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

25. Question 25: Count Vowels in a String

25.1 Code

```
import java.util.Scanner;

public class CountVowels
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

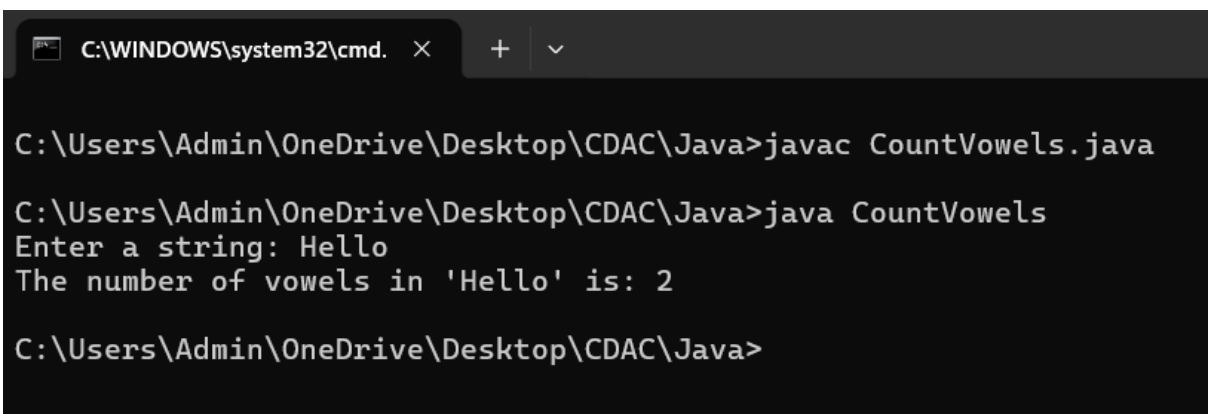
        int count = 0;

        for (int i = 0; i < str.length(); i++)
        {
            char ch = str.charAt(i);

            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
            {
                count++;
            }
        }

        System.out.println("The number of vowels in '" + str + "' is: " + count);
    }
}
```

25.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac CountVowels.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java CountVowels
Enter a string: Hello
The number of vowels in 'Hello' is: 2

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

26. Question 26: Check if a String is a Palindrome

26.1 Code

```
import java.util.Scanner;

public class StringPalindrome
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

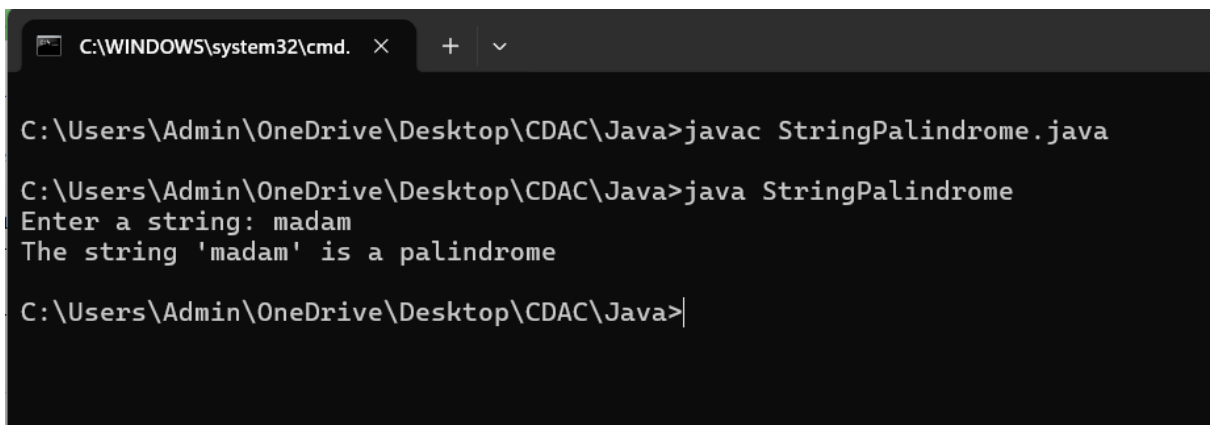
        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String reverse = "";

        for (int i = str.length() - 1; i >= 0; i--)
        {
            reverse = reverse + str.charAt(i);
        }

        if (str.equals(reverse))
        {
            System.out.println("The string '" + str + "' is a palindrome");
        }
        else
        {
            System.out.println("The string '" + str + "' is not a palindrome");
        }
    }
}
```

26.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac StringPalindrome.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java StringPalindrome
Enter a string: madam
The string 'madam' is a palindrome

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>|
```

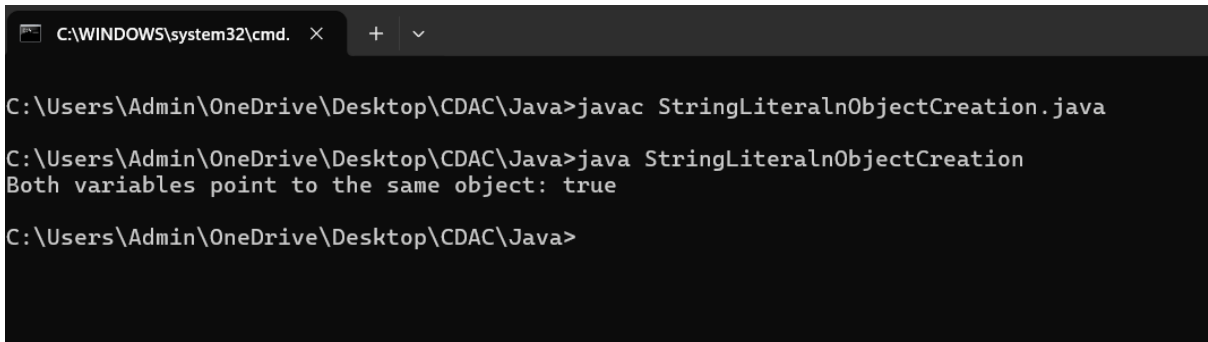
27. Question 27: String Literal and Object Creation

27.1 Code

```
public class StringLiteralObjectCreation
{
    public static void main(String args[])
    {
        String str1 = "hello";
        String str2 = "hello";

        System.out.println("Both variables point to the same object: " + (str1 ==
str2));
    }
}
```

27.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac StringLiteralObjectCreation.java
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java StringLiteralObjectCreation
Both variables point to the same object: true
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

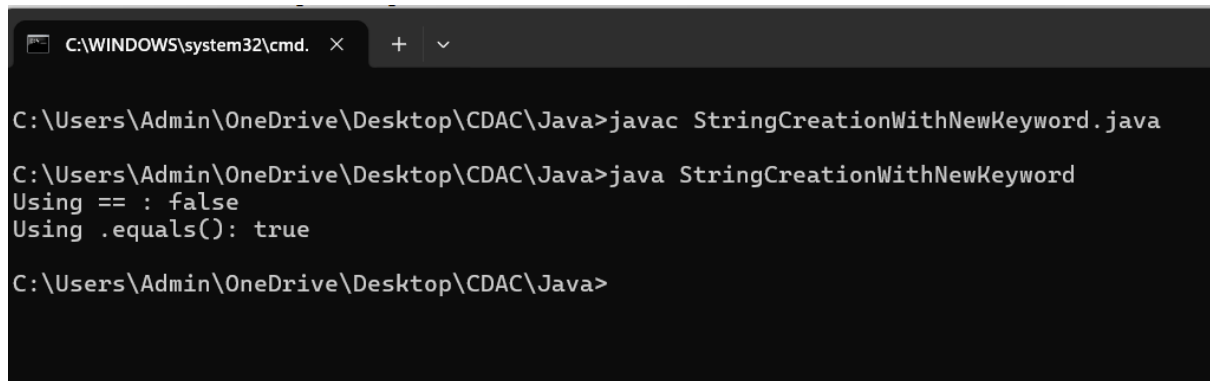
28. Question 28: String Creation with new Keyword

28.1 Code

```
public class StringCreationWithNewKeyword
{
    public static void main(String[] args)
    {
        String str1 = new String("hello");
        String str2 = new String("hello");

        System.out.println("Using == : " + (str1 == str2));
        System.out.println("Using .equals(): " + str1.equals(str2));
    }
}
```

28.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac StringCreationWithNewKeyword.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java StringCreationWithNewKeyword
Using == : false
Using .equals(): true

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

29. Question 29: String Concatenation and Object Creation

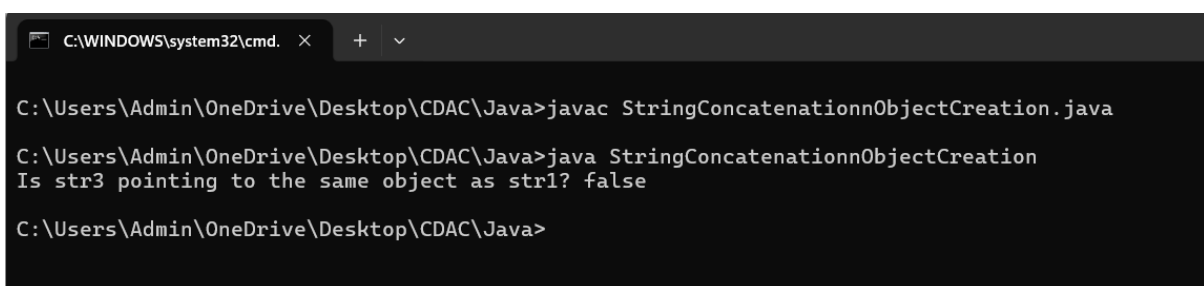
29.1 Code

```
public class StringConcatenationnnObjectCreation
{
    public static void main(String[] args)
    {
        String str1 = "hello";
        String str2 = "world";

        String str3 = str1 + str2;

        System.out.println("Is str3 pointing to the same object as str1? " + (str3 ==
str1));
    }
}
```

29.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac StringConcatenationnnObjectCreation.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java StringConcatenationnnObjectCreation
Is str3 pointing to the same object as str1? false

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

30. Question 30: String Pool with intern() Method

30.1 Code

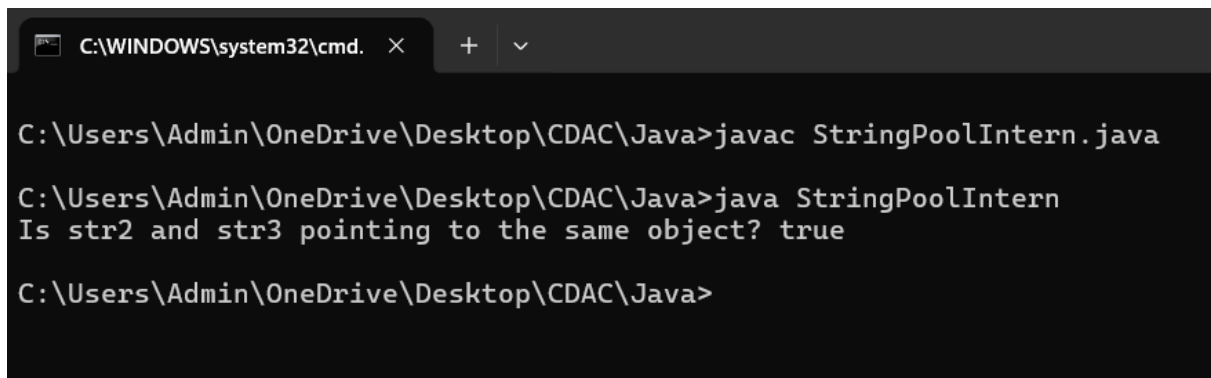
```
public class StringPoolIntern
{
    public static void main(String[] args)
    {
        String str1 = new String("hello");

        String str2 = str1.intern();

        String str3 = "hello";

        System.out.println("Is str2 and str3 pointing to the same object? " + (str2 ==
str3));
    }
}
```

30.2 Output



```
C:\WINDOWS\system32\cmd.  X  +  v

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac StringPoolIntern.java

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java StringPoolIntern
Is str2 and str3 pointing to the same object? true

C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```

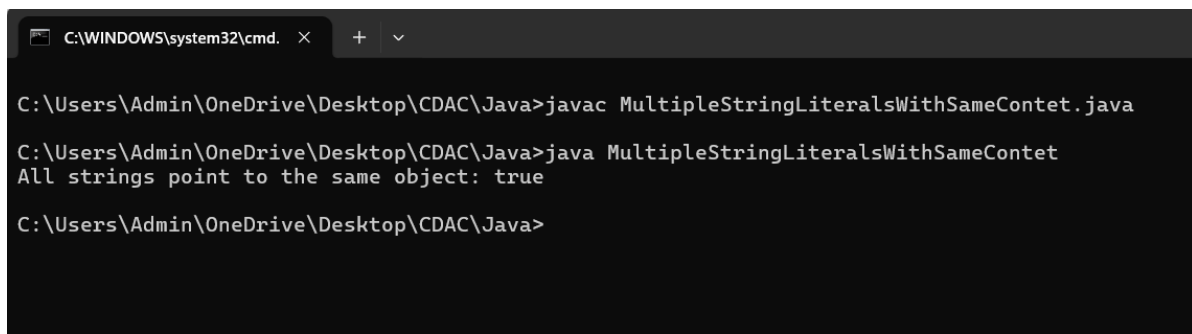
31. Question 31: Multiple String Literals with Same Content

30.1 Code

```
public class MultipleStringLiteralsWithSameContet
{
    public static void main(String[] args)
    {
        String str1 = "java";
        String str2 = "java";
        String str3 = "java";

        System.out.println("All strings point to the same object: " + (str1 == str2 &&
str2 == str3));
    }
}
```

30.1 Output



```
C:\WINDOWS\system32\cmd.  ×  +  ▾  
  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>javac MultipleStringLiteralsWithSameContet.java  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>java MultipleStringLiteralsWithSameContet  
All strings point to the same object: true  
C:\Users\Admin\OneDrive\Desktop\CDAC\Java>
```